Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

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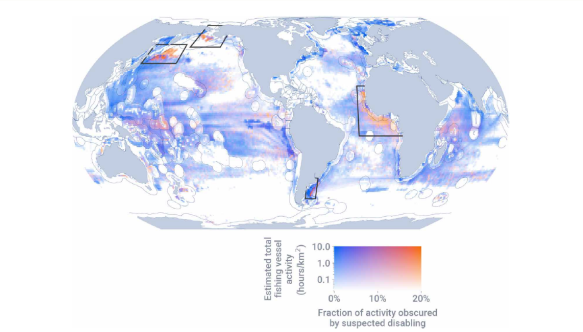
Strategic adversaries on networks



Welch et al., *Science Advances* (2022)

Credit: Global Fishing Watch

Illegal fishing, logging, trafficking.

Shared resources reached through corridors that several authorities watch at once.

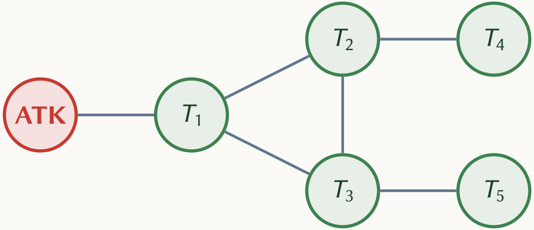
The adversary is strategic.

Routes are chosen deliberately. Where defense is weak, that is where the attack will go.

Defense is shared.

Each corridor is monitored from both

What is a network?



Nodes.

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network. Picks a target and a path.

The defenders.

One at each target. They invest to intercept.

Two traditions, one gap

Node defense (centralized)	Node defense (continuous)	Edge defense
Dziubiński & Goyal Cerdeiro et al.	**Bloch, Chatterjee & Dutta (2023)**	Washburn & Wood Mavronicolas et al.
Defense at locations. Binary, all-or-nothing. Single planner.	Defense at locations. Continuous investment. Many defenders.	Defense on connections. Discrete or budgeted. Single planner.

Bloch's model, in one slide

Survival at a node.

$$S(\text{node } i) \;=\; 1 - x_i$$

Each defender invests x_i at its own location. The attacker survives that location with probability $1 - x_i$.

Attacker's expected payoff at target *i*.

$$b_i \;\cdot \prod_{j \text{ on path to } i} (1 - x_j)$$

Target value b_i, times the probability the attacker survives every node on the way.

Equilibrium for a first target.